

UPPER HEYFORD, United Kingdom

WMO: 036553

Lat: 51.933N

Lon: 1.250W

Elev: 133

StdP: 99.74

Time Zone: 0.00 (EUW)

Period: 82-93

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.0	-3.1	-9.2	1.7	-3.9	-7.2	2.1	-1.7	14.8	8.4	13.2	7.8	3.4	60	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	25.9	17.9	24.0	17.3	22.2	16.4	18.9	23.8	17.9	22.4	17.0	21.2	3.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.4	21.1	16.1	11.6	19.7	15.2	10.9	18.8	54.5	23.8	51.2	22.4	48.2	21.2	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	9.9	8.7	-7.5	28.8	3.7	2.7	-10.1	30.8	-12.2	32.4	-14.3	33.9	-17.0	35.9		
(5)			-8.2	20.4	3.5	0.7	-10.7	21.0	-12.8	21.4	-14.8	21.8	-17.3	22.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	4.1	3.5	6.0	8.0	11.4	14.2	16.8	16.3	14.1	10.9	6.9	5.3
	DBStd	5.44	3.69	3.74	2.61	2.77	2.78	2.93	2.48	2.42	2.40	2.79	3.20	3.36
	HDD10.0	856	184	183	125	73	15	2	0	0	1	23	102	147
	HDD18.3	3138	443	417	381	312	216	129	62	72	129	232	343	403
	CDD10.0	788	0	1	2	12	58	127	210	196	123	49	8	3
	CDD18.3	28	0	0	0	0	0	4	13	9	1	0	0	0
	CDH23.3	224	0	0	0	0	4	26	121	64	8	1	0	0
	CDH26.7	33	0	0	0	0	0	1	19	13	0	0	0	0
Wind	WSAvg	4.0	4.7	4.6	4.4	4.0	3.7	3.5	3.6	3.8	3.6	3.9	4.1	4.2
Precipitation	PrecAvg	690	67	48	44	51	60	49	52	58	53	70	72	68
	PrecMax	927	180	112	88	142	141	135	143	146	113	126	124	120
	PrecMin	442	9	6	6	0	10	6	7	3	3	17	27	19
	PrecStd	127	34	28	22	35	29	32	31	35	32	30	27	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	12.9	15.2	19.2	24.0	26.0	29.0	27.9	24.2	19.8	15.0	13.2
		MCWB	10.5	10.7	10.2	13.2	16.8	18.6	18.9	19.0	17.1	15.6	13.1	11.3
	2%	DB	11.1	11.1	12.9	16.2	21.2	23.9	26.2	24.9	21.9	17.8	13.9	12.2
		MCWB	9.6	9.1	9.6	11.2	15.4	17.7	17.9	17.6	16.1	14.7	12.3	10.4
	5%	DB	10.2	10.0	11.1	14.3	18.9	21.9	24.2	23.0	20.0	16.2	12.9	11.2
		MCWB	8.8	8.4	8.6	10.3	13.7	16.7	17.3	16.8	15.5	13.6	11.2	9.5
	10%	DB	9.2	8.9	10.1	13.0	16.9	19.9	22.2	21.2	18.2	15.1	11.9	10.2
		MCWB	8.0	7.7	8.2	9.5	12.6	15.2	16.4	16.0	14.3	12.7	10.4	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	10.9	11.4	14.1	17.7	19.6	20.5	19.8	18.1	16.1	13.4	11.9
		MCDB	12.2	12.5	14.4	17.6	22.8	24.1	26.9	25.6	22.8	18.8	14.5	13.0
	2%	WB	10.1	9.7	10.3	12.2	15.7	18.0	19.0	18.6	16.8	15.1	12.2	11.1
		MCDB	11.0	10.8	11.9	15.4	20.1	22.8	24.8	23.1	20.3	17.2	13.5	12.5
	5%	WB	9.2	8.6	9.4	11.0	14.3	16.7	18.0	17.5	16.1	14.1	11.6	10.2
		MCDB	10.1	9.8	10.5	13.5	18.2	21.0	22.6	21.6	19.2	15.9	12.8	11.4
	10%	WB	8.1	7.5	8.4	10.0	13.0	15.6	16.9	16.6	15.2	13.0	10.6	9.1
		MCDB	9.3	8.7	9.8	12.5	16.2	19.4	21.4	20.3	17.8	14.9	11.9	10.4
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.2	6.3	7.6	8.4	8.2	9.0	8.7	7.6	6.2	5.0	4.4
		MCDBR	5.7	6.1	7.3	9.7	11.9	11.2	12.6	11.2	10.1	6.9	5.2	5.0
		MCWBR	5.8	5.4	5.3	6.2	7.1	6.4	6.1	5.6	5.8	4.9	4.7	4.8
	5% WB	MCDBR	5.9	5.7	6.5	9.0	11.0	10.6	10.9	10.1	8.8	6.3	5.4	4.9
		MCWBR	6.1	5.2	5.4	6.3	7.1	6.4	6.1	5.7	5.9	5.1	5.1	5.0
Clear-Sky Solar Irradiance	taub	0.320	0.336	0.369	0.393	0.398	0.396	0.393	0.382	0.365	0.349	0.323	0.310	
	taud	2.368	2.385	2.287	2.233	2.255	2.285	2.301	2.355	2.402	2.433	2.414	2.349	
	Ebn at Noon	680	780	817	840	854	858	853	844	816	755	673	622	
	Edh at Noon	62	79	105	125	128	126	122	109	93	75	58	53	
All-Sky Solar Radiation	RadAvg	0.79	1.46	2.54	3.90	4.76	5.19	4.95	4.10	3.09	1.80	0.95	0.63	
	RadStd	0.07	0.18	0.33	0.44	0.30	0.52	0.51	0.32	0.27	0.16	0.11	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (95 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page